

## Readme

All the files needed for this unit are in this folder. The first document you should look at is

### Unit 3 Teaching Guide with Answers.docx

This unit covers both the topics of Computer architecture and data storage. The first lesson considers CPU architecture including key components such as the ALU and Control Unit and key registers such as the PC, MAR, MDR, CIR and accumulator. In the second lesson, the factors that affect CPU performance are covered. Lessons 3 and 4 cover embedded systems, instruction sets and memory. Finally, secondary storage, virtual memory and cloud storage are all considered in the final two lessons. A careful consideration of secondary storage includes an internal understanding of how key storage technologies work, including the use of platters, tracks, sectors and electromagnets for magnetic storage, pits and lands for optical storage and NAND or NOR gates for solid-state storage. The final lesson considers cloud storage along with its advantages and disadvantages compared to local storage. At the end of the unit a final assessment test is given which covers all the lessons in the unit.

You may wish to copy the files from the unit onto a secure shared network drive or VLE where students can access the presentations, worksheets, homeworks and links. Remember to remove any documents you would prefer them not to access such as the assessment and answer sheets.

You may also wish to upload the files for secure home access. This could be via an MLE such as Fronter, Frog, SharePoint or Moodle.



If you have any difficulty opening any of the files, which were all created in Office 365, or experience any other problems, please email [support@pgonline.co.uk](mailto:support@pgonline.co.uk).

This is the third in a series of 10 units for Cambridge IGCSE 0478/0984 and O Level 2210 Computer Science. We hope you will find it useful.

We would greatly appreciate your feedback once you have had a chance to use the material. Please send an email to [support@pgonline.co.uk](mailto:support@pgonline.co.uk).

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